

AMRO TABCHE

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PROFESSIONAL SUMMARY

Mechanical Engineering graduate (University of Ottawa, 2026) with design–build–test experience spanning an industry capstone — a two-axis electromechanical platform delivered to ING Robotic Aviation — a manufacturing co-op in telecom PCB production, and a Design Day–winning assistive device. Honeywell-certified on Tridium Niagara and Alerton building-automation platforms. Canadian citizen.

TECHNICAL SKILLS

CAD / Simulation: SolidWorks, MATLAB/Simulink, FEA, structural & parametric optimization, GD&T

Controls / Automation: Tridium Niagara 4, Alerton Compass, JACE 8000, BACnet (IP/MS-TP), Modbus RS-485, DDC, HVAC/VAV, ASHRAE standards

Manufacturing / Test: Prototyping, 3D printing, DFM, ICT (bed-of-nails), 4-wire Kelvin measurement, MIG welding, sheet metal

Programming / Tools: Python, Arduino (C/C++), embedded systems, Git, LaTeX, Microsoft Office

Languages: English (Fluent), Arabic (Native), French (Beginner)

WORK EXPERIENCE

Building Management Systems Engineering Training

Aug 2023; Jul – Aug 2024

Honeywell

Riyadh, Saudi Arabia

- Configured Alerton BACnet VAV HVAC controllers and Tridium JACE 8000 DDC supervisors on live equipment at the Sofitel Riyadh, integrating BACnet IP/MS-TP and Modbus RS-485 protocols.
- Programmed control logic and scheduling sequences in Alerton Compass; wired and mapped Modbus RS-485 networks (polling, register mapping).
- Performed system integration and troubleshooting across BAS architectures and produced structured inspection reports as certification deliverables.

Mechanical Engineering Co-op

Jan 2024 – Apr 2024

Comtest Networks Inc.

Nepean, ON

- Ran prototyping trials and process-optimization initiatives for telecom-sector PCB production, spanning design review through assembly.
- Performed in-circuit testing (ICT) of populated PCB assemblies using bed-of-nails fixtures and 4-wire Kelvin measurement at milliohm precision, catching shorts, opens, and continuity faults before downstream assembly.
- Maintained BOMs and MRP data for production scheduling and inventory accuracy; implemented ESD-safe handling procedures across work cells.

PROJECTS

Two-Axis Pan/Tilt Turret System — Industry Capstone

Sep 2025 – Apr 2026

ING Robotic Aviation Inc.

Ottawa, ON

- Designed a two-axis pan/tilt electromechanical system (counter-drone defensive platform) achieving 360° continuous azimuth and 90° elevation under strict size, mass, and power constraints, integrating servo-motor and harmonic-drive actuation and a thermal-imaging payload.
- Engineered a recoil-management mechanism — modeling recoil dynamics as a spring–damper ODE in MATLAB to size commercial industrial dampers — and an all-weather sealed enclosure preserving structural integrity under operational loading.
- Performed FEA-based structural analysis and parametric optimization of three critical components (mounting plate, carriage-plate damper interface, guide rods) against strength, recoil response, mass, and drag.
- Built a MATLAB optimization tool linked directly to the SolidWorks CAD assembly, enabling automated geometry updates and real-time validation of candidate configurations.

One-Handed Pet Waste Scooping Device — Design Day Winner

May 2025 – Jul 2025

Accessible Design Category

Ottawa, ON

- Designed and built a 380 g assistive device enabling a quadriplegic wheelchair user to retrieve dog waste single-handedly, integrating a servo-driven claw, Arduino control, magnetic bag attachment, and a carbon-fiber shaft.
- Ran 7+ structured client-testing sessions across grip force, reach, bag-removal time, and terrain adaptability; iterated claw geometry and wrist-strap sizing from direct user feedback.
- Delivered a complete documentation package: user manual, BOM (~\$45 prototype cost), and assembly instructions.

Two-Wheeled Voice-Controlled Mechatronic Platform

Oct 2023 – Dec 2023

- Built an autonomous two-wheeled platform on Arduino UNO with multi-modal behaviors: speech-recognition drive commands (forward/reverse/turn/speed), ultrasonic obstacle detection with automatic motor cutoff under 10 cm, and photoresistor-triggered headlights.
- Modeled a three-channel sensor signal-conditioning chain in Simulink (instrumentation amplification, low-pass filtering, multiplexed sample-and-hold, 12-bit ADC) and designed the chassis in SolidWorks with a mass-tuned lower compartment for stability.

EDUCATION & CERTIFICATIONS

University of Ottawa — BAsc, Mechanical Engineering · Apr 2026

Honeywell-certified: Tridium Niagara 4.13 (Networking, JACE 8000, BACnet/Modbus); Alerton (VLC, BACnet IP/MS-TP, Compass)